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(57) Abstract:

The automatic water purification system (100) comprises a raw water inlet (102) to receive natural water, a screening unit (104) to remove suspended impurities, and a plurality of sensors (106) that measure pH, total dissolved solids, and turbidity. A microprocessor-based controller (108) processes this sensor data to classify the water type and activate a corresponding purification sequence through a plurality of purification stages (110) before delivery via a purified water outlet (112). Additionally, the system features an electro-mineral conditioning module (114) with low-voltage DC-energized iron (116) and silver (118) electrodes for mineral enrichment and antibacterial action. It also comprises a pH stabilization module (120) using a calcium-magnesium alloy (122) for chemical-free potable pH adjustment, and a salt removal module (124) utilizing a porous ceramic disk (126) with reverse osmosis membranes (128) attached to its upper and lower surfaces. (FIGURE 1)

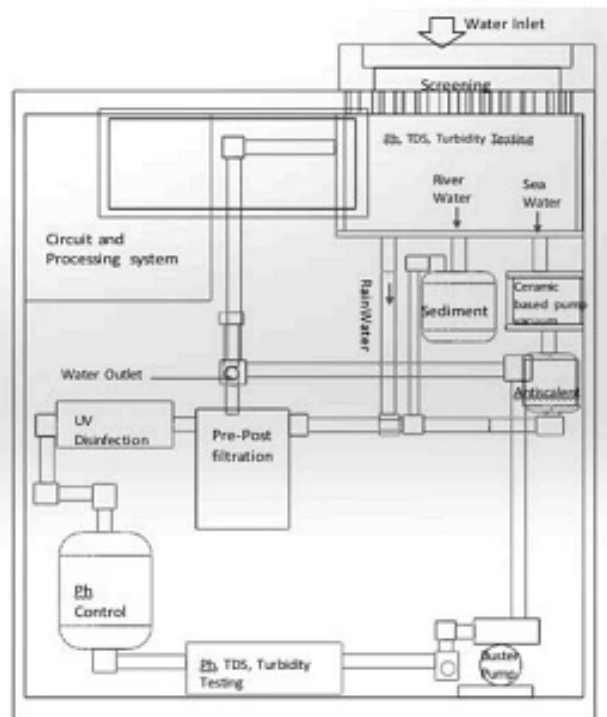


FIGURE 1

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

1. TITLE OF THE INVENTION

AUTOMATIC WATER PURIFICATION SYSTEM AND METHOD FOR
PURIFYING WATER FROM MULTIPLE NATURAL WATER SOURCES

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3. PREAMBLE TO THE DESCRIPTION

The following complete specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF INVENTION

[0001] The present invention relates to the field of water purification and desalination systems, particularly to automated water treatment systems capable of purifying multiple natural water sources. More particularly the invention relates to a system and method of automated water treatment systems through microprocessor-controlled water purification system using sensor-based classification, electro-mineral conditioning, alloy-based pH stabilization, and ceramic-membrane assisted salt removal techniques. The present invention provides reduced maintenance cost, lower power consumption, and simplified operational complexity compared to conventional purification systems.

BACKGROUND OF THE INVENTION

[0002] Water purification systems are used to remove impurities, contaminants, and dissolved substances from water to produce water suitable for drinking and domestic use. Conventional water purification systems employ various techniques comprising filtration, reverse osmosis, ultraviolet disinfection, and chemical treatment to achieve desired water quality standards. These systems are typically designed to process water from specific sources such as municipal water supplies, groundwater, or surface water, with each system configured for the particular characteristics of the intended water source.

[0003] Existing water purification systems face several limitations and challenges. Conventional systems are typically designed for specific water sources and lack adaptability to process water from multiple natural sources with varying quality parameters. Most purifiers rely on cartridge-based mineralization requiring frequent replacement, chemical pH control requiring continuous dosing, or energy-intensive distillation methods for salt removal. These approaches increase maintenance cost, power consumption, and operational complexity. Furthermore, most purifiers cannot automatically classify water quality or purify high-salinity water efficiently using low-power mechanisms.

[0004] There are several patents related to the invention. In one such application, US Patent No. 6,428,705 discloses a water purification system using multiple filtration stages and chemical treatment for removing contaminants from water. The system employs cartridge-based filters that require periodic replacement and chemical dosing for pH adjustment. However, this system has drawbacks comprising high maintenance requirements due to frequent cartridge replacement, dependency on chemical additives for pH control, and inability to automatically adapt to different water source types.

[0005] In another patent application, US Patent No. 8,147,697 discloses a desalination system using thermal distillation for removing dissolved salts from seawater. The system heats water to

produce steam which is then condensed to obtain purified water. However, this system has drawbacks comprising high energy consumption due to the thermal distillation process, complex equipment requirements, and unsuitability for low-power applications in remote or off-grid locations.

[0006] In view of the above limitations, there is a need for an automatic water purification system capable of purifying water from multiple natural water sources using a single integrated architecture, eliminating cartridge-based mineralization through electro-mineral conditioning, maintaining pH naturally using alloy-based stabilization without chemical dosing, and removing dissolved salts using a non-thermal low-energy mechanism.

OBJECTS OF THE INVENTION

[0007] The primary objective of the present invention is to provide an automatic water purification system capable of purifying water from multiple natural water sources comprising rain water, river water, pond water, ground water, sea water, and ocean water using a single integrated architecture.

[0008] Another objective of the present invention is to provide an automatic water purification system that eliminates cartridge-based mineralization through electro-mineral conditioning using iron and silver electrodes energized with low-voltage DC power.

[0009] Another objective of the present invention is to provide an automatic water purification system that maintains water pH within a potable range using alloy-based stabilization without requiring chemical dosing.

[0010] Another objective of the present invention is to provide an automatic water purification system that removes dissolved salts using a non-thermal low-energy mechanism employing a porous ceramic disk with reverse osmosis membranes.

[0011] Another objective of the present invention is to provide an automatic water purification system with sensor-based water quality measurement and microprocessor-controlled classification for automatic selection of purification sequences corresponding to different water types.

[0012] Another objective of the present invention is to provide an automatic water purification system with reduced maintenance cost by eliminating frequently replaceable components such as mineral cartridges and chemical additives.

[0013] Another objective of the present invention is to provide an automatic water purification system with lower power consumption suitable for household-level and small community-level applications comprising remote or off-grid locations.

[0014] Yet another objective of the present invention is to provide an automatic water purification system with simplified operational complexity requiring minimal human intervention through fully automatic operation.

SUMMARY OF THE INVENTION

5 [0015] The present invention provides an automatic water purification system and method for treating water from multiple natural sources. The system includes a raw water inlet, a screening unit for suspended impurities, a purified water outlet, and multiple sensors that measure pH, total dissolved solids, and turbidity. A microprocessor-based controller analyzes this sensor data to classify the water type and activate a corresponding purification sequence. The system's purification stages feature an
10 electro-mineral conditioning module with iron and silver electrodes for mineral enrichment and antibacterial action, a chemical-free pH stabilization module utilizing a calcium-magnesium alloy, and a salt removal module composed of a porous ceramic disk with reverse osmosis membranes attached to its upper and lower surfaces.

[0016] In some aspects, the salt removal module comprises protective mesh layers provided on
15 outer sides of the reverse osmosis membranes to prevent mechanical damage. The salt removal module comprises a pump configured to generate a localized pressure-induced vacuum effect to assist in salt separation during water flow through the porous ceramic disk. The iron electrode and the silver electrode of the electro-mineral conditioning module arranged as layered mineral plates positioned on opposite sides of a central flow chamber. The pH stabilization module include a calcium electrode and
20 a magnesium electrode positioned on opposite sides of a central flow chamber. The system comprises a UV disinfection unit configured to neutralize microorganisms in the water. The microprocessor-based controller configured to perform a final quality check of the water and redirect the water back to the raw water inlet when measured water quality parameters do not fall within acceptable drinking water parameters. For water classified as sea water or ocean water, the method comprises passing the
25 water through the salt removal module. The present invention provides a single integrated architecture enabling purification of all natural water sources while reducing maintenance requirements and operational costs.

BRIEF DESCRIPTION OF FIGURES

[0017] The present invention will be better understood after reading the following detailed
30 description of the presently preferred aspects thereof with reference to the appended drawings, in

which the features, other aspects and advantages of certain exemplary embodiments of the invention will be more apparent from the accompanying drawing in which:

[0018] FIGURE 1 FIGURE 1 illustrates a process flow diagram of an automatic smart water purification system, according to aspects of the present disclosure.

5 **[0019]** FIGURE 2 illustrates multiple views of a salt removal module used in the water purification system, according to aspects of the present disclosure.

[0020] Fig. 2(a) shows an exploded view of the salt removal module, according to an embodiment.

10 **[0021]** Fig. 2(b) presents an assembled isometric view of the salt removal module, according to an embodiment.

[0022] Fig. 2(c) provides a labeled exploded view of the salt removal module, according to an embodiment.

[0023] FIGURE 3 illustrates multiple views of modular components used in the water purification system, according to aspects of the present disclosure.

15 **[0024]** Fig. 3(a) shows an isometric view of a cylindrical filter assembly, according to an embodiment.

[0025] Fig. 3(b) depicts an isometric view of a rectangular mesh or membrane panel, according to an embodiment.

[0026] Fig. 3(c) presents exploded views of treatment modules, according to an embodiment.

20 **[0027]** FIGURE 4 illustrates a circuit connection diagram of the automatic water purification system, according to aspects of the present invention.

[0028] FIGURE 5 illustrates a flowchart for a water purification process, according to aspects of the present invention.

DETAILED DESCRIPTION

25 **[0029]** The following detailed description and embodiments set forth herein below are merely exemplary out of the wide variety and arrangement of instructions which be employed with the present

invention. The present invention embodied in other specific forms without departing from the spirit or essential characteristics thereof. All the features disclosed in this specification replaced by similar other or alternative features performing similar or same or equivalent purposes. Thus, unless stated otherwise, they all are within the scope of the present invention.

5 **[0030]** Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the embodiments described herein be made without departing from the scope of the invention. In addition, descriptions of well-known functions and constructions are omitted for clarity and conciseness.

10 **[0031]** The terms and words used in the following description and claims are not limited to the bibliographical meanings but are merely used to enable a clear and consistent understanding of the invention. Accordingly, it should be apparent to those skilled in the art that the following description of exemplary embodiments of the present invention are provided for illustration purpose only and not for the purpose of limiting the invention.

15 **[0032]** It is to be understood that the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise.

[0033] It should be emphasized that the term "comprises/comprising" when used in this specification is taken to specify the presence of stated features, integers, steps, or components but does not preclude the presence or addition of one or more other features, integers, steps, components, or groups thereof.

20 **[0034]** Accordingly, the present invention relates to the field of smart water purification and desalination systems. The present invention provides an automatic water purification system (100) that is microprocessor-controlled and capable of purifying multiple natural water sources. The system (100) uses sensor-based classification to identify water types, electro-mineral conditioning to provide mineral enrichment and antibacterial action, alloy-based pH stabilization to maintain water pH within
25 a potable range, and ceramic-membrane assisted salt removal techniques to remove dissolved salts. The system (100) operates within the domains of environmental engineering, water resource management, and low-power automated purification technologies.

[0035] The automatic water purification system (100) for purifying water from multiple natural water sources comprises a raw water inlet (102) configured to receive water from a natural water

source, a screening unit (104) configured to remove suspended impurities from the water, a plurality of sensors (106) configured to measure water quality parameters comprising pH, total dissolved solids, and turbidity, and a microprocessor-based controller (108) configured to receive sensor data, classify a water type based on the sensor data, and activate a purification sequence corresponding to the classified water type. The system (100) comprises a plurality of purification stages (110) and a purified water outlet (112). The system (100) comprises an electro-mineral conditioning module (114) with an iron electrode (116) and a silver electrode (118), a pH stabilization module (120) with a calcium-magnesium alloy (122), and a salt removal module (124) with a porous ceramic disk (126) and reverse osmosis membranes (128). The system (100) eliminates cartridge-based mineralization, chemical pH control, and energy-intensive distillation methods, thereby reducing maintenance cost, power consumption, and operational complexity.

[0036] In the present invention, FIGURE 1 illustrates a process flow diagram of an automatic smart water purification system. The system includes a water inlet at the top through which raw water from various natural sources enters the system. The water first passes through a screening unit where large suspended impurities such as leaves, plastics, sand particles, and debris are removed to protect downstream components.

[0037] After screening, the water flows to a pH, TDS, and turbidity testing zone where sensors continuously measure the quality of the incoming water. A pH sensor measures the acidity or alkalinity of the water, a TDS sensor measures total dissolved solids, and a turbidity sensor detects the clarity and suspended particle content of the water. These sensor values are transmitted to a circuit and processing system positioned on the left side of the diagram, which analyzes the data and automatically classifies the type of water source.

[0038] Based on the classified water type, the microprocessor-based control unit activates the appropriate purification sequence by operating DC pump motors and solenoid control valves. The system directs water through different treatment paths depending on the water source classification. For river water and sea water, separate inlet paths are provided. River water flows through a sediment treatment stage, while sea water is directed to a ceramic based pump vacuum unit for salt removal processing. An antiscalent unit is positioned downstream to remove residual dissolved ions and hardness-causing minerals.

[0039] A rainwater path is also provided, which connects to the main processing line. The system includes a pre-post filtration unit that handles filtration requirements for the various water sources. A UV disinfection unit is positioned to neutralize harmful microorganisms such as bacteria and viruses.

[0040] The circuit and processing system manages sensor inputs, mineral activation, flow routing, and purification sequencing. A pH control module is connected to the processing system to maintain water pH within the potable range. After pH control, the water passes through another pH, TDS, and turbidity testing stage to verify final water quality.

5 **[0041]** A booster pump is positioned at the bottom right of the system to maintain adequate water pressure throughout the purification process. The water outlet is located on the left side of the diagram, through which purified water exits the system ready for safe drinking or domestic use. The entire purification cycle operates automatically with minimal human intervention, utilizing a microprocessor-controlled architecture that enables adaptive operation based on varying input water
10 conditions.

[0042] FIGURE 2 illustrates multiple views of a salt removal module used in the water purification system. For sea or ocean water with high TDS, water is first directed to the salt removal module before other treatment stages. Fig. 2(a) shows an exploded view of the salt removal module, depicting the layered arrangement of components including protective mesh layers, membrane
15 elements, a porous ceramic disk, and associated structural supports. The exploded view reveals how the individual filtration layers are stacked and assembled to form the complete module.

[0043] Fig. 2(b) presents an assembled isometric view of the salt removal module, showing the integrated unit with water inlet and outlet connections positioned on the cylindrical body. The assembled view demonstrates the compact configuration of the module with the filtration components
20 housed within a central chamber.

[0044] Fig. 2(c) provides a labeled exploded view identifying the specific components of the salt removal assembly. A water inlet is positioned at the top of the module, followed by a protective mesh layer, an RO membrane, a porous ceramic disk at the center, another RO membrane below the ceramic disk, and another protective mesh layer. A water outlet is positioned on the side of the assembly. A
25 pump to create vacuum is connected at the bottom of the module. The porous ceramic disk has RO membranes attached on both upper and lower sides with protective mesh layers on both sides of the membranes. The protective mesh layers prevent membrane rupture and mechanical damage under pressure conditions.

[0045] In an exemplary embodiment, downward water flow through the porous ceramic disk
30 generates a localized pressure-induced vacuum effect for salt separation without thermal distillation or high-pressure pumps. The layered configuration enables salt separation through the pressure-induced localized vacuum effect generated during controlled water flow through the ceramic disk.

Accumulated salt deposits on the membrane surface are naturally detached through reverse flow action providing self-cleaning capability. When flow conditions change, the reverse flow action naturally detaches accumulated salt deposits, reducing fouling and extending membrane life.

[0046] FIGURE 3 illustrates multiple views of modular components used in the water purification system. Fig. 3(a) shows an isometric view of a cylindrical filter assembly comprising a cylindrical housing with a helical coil element wrapped around its exterior surface. The assembly includes circular end caps positioned at both ends of the cylindrical housing, with a lateral inlet or outlet tube extending from the side of the structure. This configuration facilitates water flow through the cylindrical chamber while the helical element provides contact surface area for treatment processes.

[0047] Fig. 3(b) depicts an isometric view of a rectangular mesh or membrane panel. The panel features a flat rectangular frame with a grid pattern visible on its surface, indicating a porous or permeable structure. This component serves as a filtration or separation element within the purification system.

[0048] Fig. 3(c) presents two exploded views of treatment modules. The upper exploded view shows a pH control module comprising a Cu electrode positioned at the top, a central treatment chamber with water inlet and water outlet ports extending laterally, and an Mg electrode positioned at the bottom. The electrodes are arranged on opposite sides of the chamber to facilitate electrochemical pH adjustment of water flowing through the module. The calcium-magnesium alloy module naturally adjusts and maintains water pH within the potable range without chemical dosing.

[0049] The lower exploded view illustrates a mineral adding module with a similar configuration, featuring an Fe electrode at the top position, a central treatment chamber with water inlet and water outlet connections, and an Ag electrode at the bottom position. This arrangement enables controlled release of iron and silver ions into the water stream for mineral conditioning and antibacterial treatment. Both modules utilize a layered electrode configuration that allows water to flow between the electrode surfaces during treatment.

[0050] In an exemplary embodiment, the electro-mineral conditioning module uses iron and silver layered mineral plates energized with low-voltage DC power of 3-5 V. The iron and silver layered plates provide controlled mineral enrichment along with antibacterial action without using mineral cartridges. The low-voltage DC power operation at 3-5 V enables controlled mineral release and antibacterial action without consumable cartridges. The passive pH adjustment using the calcium-magnesium alloy maintains water pH within the potable range without chemical dosing, eliminating the need for active electronic pH controllers or chemical additives.

[0051] FIGURE 4. The plurality of sensors (106) is configured to measure water quality parameters comprising pH, total dissolved solids, and turbidity. The microprocessor-based controller (108) is configured to receive sensor data from the plurality of sensors (106), classify a water type based on the sensor data, and activate a purification sequence corresponding to the classified water type. FIGURE 4 shows a circuit connection diagram of the system (100). The circuit connection diagram comprises a Raspberry Pi 5 microprocessor serving as the microprocessor-based controller (108), an ADS1115 analog-to-digital converter module connected to the Raspberry Pi 5 for reading analog sensor signals, a pressure sensor, a TDS sensor, a water level sensor, a temperature sensor, a turbidity sensor, and a pH sensor. The Raspberry Pi 5 interfaces with an 8 channel relay module that manages switching of actuators comprising a 12V solenoid valve, a water pump, and a UV disinfection unit.

[0052] In an exemplary embodiment, the system (100) comprises DC pump motors that are activated by the microprocessor-based controller (108) based on the classified water type. The system (100) comprises solenoid control valves that are operated by the microprocessor-based controller (108) to route water through appropriate purification stages. The microprocessor-based controller (108) analyzes sensor data to classify the water type and activates appropriate purification sequences by controlling the pump motors and solenoid valves through the relay module.

[0053] FIGURE 5 shows a flowchart for the water purification process. The process begins at an Inlet stage where raw water from a natural source enters the system (100). The water passes through a Screening step where large suspended impurities are removed. The water then flows to a pH, TDS and Turbidity Testing step where the plurality of sensors (106) continuously measure the quality of the incoming water. Based on the measurements, the microprocessor-based controller (108) classifies the water into one of three categories.

[0054] For Sea/Ocean water, the process proceeds through a Salt Removal step followed by an Anti-scalant step where residual dissolved ions and hardness-causing minerals are removed. For Pond/River Water, the process comprises a Prefiltration step followed by a Sedimentation step where heavier suspended solids settle and are removed. For Rain/Ground Water, the water proceeds directly to the common purification stages.

[0055] All three water paths converge at a Pre Carbon Filter step, followed by a Post Carbon Filter step. The water then passes through a Reverse Osmosis step for membrane-based filtration. The water undergoes UV Disinfection to neutralize harmful microorganisms, followed by an Ultra Filtration step to remove remaining fine particles. The process continues to a pH control step using the calcium-

magnesium alloy (122) in the pH stabilization module (120). The water then passes through an Electro Mineral Conditioning step using the iron electrode (116) and silver electrode (118) in the electro-mineral conditioning module (114).

[0056] After conditioning, the water undergoes Final pH, TDS, Turbidity Testing. If the measured values do not fall within acceptable drinking water parameters, the process redirects the water back to the Screening step and the purification process is repeated. If the output water meets the required quality standards, the water proceeds to the Clean Water Outlet for safe drinking or domestic use.

[0057] In an embodiment, as shown in FIGURE 5, a method for purifying water from multiple natural water sources comprises the following steps:

[0058] receiving water from a natural water source through a raw water inlet (102);

[0059] screening the water to remove suspended impurities;

[0060] measuring water quality parameters comprising pH, total dissolved solids, and turbidity using a plurality of sensors (106);

[0061] classifying a water type based on the measured water quality parameters using a microprocessor-based controller (108);

[0062] activating a purification sequence corresponding to the classified water type;

[0063] passing the water through an electro-mineral conditioning module (114) comprising an iron electrode (116) and a silver electrode (118) energized with low-voltage DC power to provide controlled mineral enrichment and antibacterial action;

[0064] passing the water through a pH stabilization module (120) comprising a calcium-magnesium alloy (122) to adjust and maintain water pH within a potable range without chemical dosing;

[0065] for water classified as sea water or ocean water, passing the water through a salt removal module (124) comprising a porous ceramic disk (126) with reverse osmosis membranes (128) attached on upper and lower surfaces of the porous ceramic disk (126);

[0066] performing a final quality check of the water and redirecting the water back to the raw water inlet (102) when measured water quality parameters do not fall within acceptable drinking water parameters; and

[0067] outputting purified water through a purified water outlet (112).

[0068] The present invention provides the following advantages:

- The present invention provides reduction in maintenance by eliminating mineral cartridges, chemical additives, and frequently replaceable components.
- The present invention provides low voltage and low power operation making the system (100) more energy-efficient than thermal distillation and high-pressure desalination systems.
- The present invention provides effective salt removal using the porous ceramic disk (126) combined with the reverse osmosis membranes (128) and the protective mesh layers (130) while reducing system complexity and improving component life.
- The present invention provides automatic water quality detection, classification, and process selection ensuring safe and consistent drinking water under varying input conditions.
- The present invention provides a non-thermal and non-chemical purification approach making the system (100) environmentally safe and suitable for long-term operation.
- The present invention provides a single integrated architecture enabling purification of all natural water sources, reducing cost and eliminating the need for multiple dedicated systems.

[0069] Features of any of the examples or embodiments outlined above combined to create additional examples or embodiments without losing the intended effect. It should be understood that the description of an embodiment or example provided above is by way of example only, and various modifications could be made by one skilled in the art. Furthermore, one skilled in the art will recognise that numerous modifications and combinations of various aspects are possible. Accordingly, the described aspects are intended to encompass all such alterations, modifications, and variations that fall within the scope of the appended claims.

We Claim:

1. An automatic water purification system (100) for purifying water from multiple natural water sources, the system (100) comprising:

- a. a raw water inlet (102) configured to receive water from a natural water source;
- b. a screening unit (104) fluidly connected to the raw water inlet (102) and configured to remove suspended impurities from the water;
- c. a plurality of sensors (106) fluidly connected downstream of the screening unit (104) and configured to measure water quality parameters comprising pH, total dissolved solids, and turbidity;
- d. a microprocessor-based controller (108) connected to the plurality of sensors (106) and configured to receive sensor data, classify a water type based on the sensor data, and activate a purification sequence corresponding to the classified water type;
- e. a plurality of purification stages (110) fluidly connected downstream of the plurality of sensors (106);
- f. a UV disinfection unit (140) fluidly connected to the plurality of purification stages (110) and configured to neutralize microorganisms in the water; and
- g. a purified water outlet (112) fluidly connected downstream of the UV disinfection unit (140);

wherein,

- wherein, a plurality of purification stages (110) comprises:
 - i. an electro-mineral conditioning module (114) comprising an iron electrode (116) and a silver electrode (118) configured to be energized with low-voltage DC power to provide controlled mineral enrichment and antibacterial action;
 - ii. a pH stabilization module (120) comprising a calcium-magnesium alloy (122) configured to adjust and maintain water pH within a potable range without chemical dosing; and
 - iii. a salt removal module (124) comprising a porous ceramic disk (126), wherein reverse osmosis membranes (128) are attached on upper and lower surfaces of the porous ceramic disk (126).

- the salt removal module (124) is configured such that downward water flow through the porous ceramic disk (126) generates a localized pressure-induced vacuum effect to separate dissolved salts without thermal distillation or high-pressure pumps; and
 - the reverse osmosis membranes (128) are configured for self-cleaning action, wherein accumulated salt deposits are naturally detached when flow conditions change.
- 5
2. The system (100) as claimed in claim 1, wherein the salt removal module (124) comprises protective mesh layers (130) provided on outer sides of the reverse osmosis membranes (128) to prevent mechanical damage.
 3. The system (100) as claimed in claim 1, wherein the salt removal module (124) comprises a pump (132) configured to generate a localized pressure-induced vacuum effect to assist in salt separation during water flow through the porous ceramic disk (126).
 - 10 4. The system (100) as claimed in claim 1, wherein the iron electrode (116) and the silver electrode (118) of the electro-mineral conditioning module (114) are arranged as layered mineral plates positioned on opposite sides of a central flow chamber (134).
 - 15 5. The system (100) as claimed in claim 1, wherein the pH stabilization module (120) comprises a calcium electrode (136) and a magnesium electrode (138) positioned on opposite sides of a central flow chamber (134).
 6. The system (100) as claimed in claim 1, wherein comprising a UV disinfection unit (140) configured to neutralize microorganisms in the water.
 - 20 7. The system (100) as claimed in claim 1, wherein the microprocessor-based controller (108) is configured to perform a final quality check of the water and redirect the water back to the raw water inlet (102) when measured water quality parameters do not fall within acceptable drinking water parameters.
 - 25 8. A method (200) for purifying water from multiple natural water sources, the method comprising:
 - a. receiving water from a natural water source through a raw water inlet (102);
 - b. screening the water to remove suspended impurities;
 - c. measuring water quality parameters comprising pH, total dissolved solids, and turbidity using a plurality of sensors (106);

- d. classifying a water type based on the measured water quality parameters using a microprocessor-based controller (108);
- e. activating a purification sequence corresponding to the classified water type;
- f. passing the water through an electro-mineral conditioning module (114) comprising an iron electrode (116) and a silver electrode (118) energized with low-voltage DC power to provide controlled mineral enrichment and antibacterial action;
- g. passing the water through a pH stabilization module (120) comprising a calcium-magnesium alloy (122) to adjust and maintain water pH within a potable range without chemical dosing; and
- h. outputting purified water through a purified water outlet (112).

9. The method (200) as claimed in claim 8, wherein comprising, for water classified as sea water or ocean water, passing the water through a salt removal module (124) comprising a porous ceramic disk (126) with reverse osmosis membranes (128) attached on upper and lower surfaces of the porous ceramic disk (126).

10. The method (200) as claimed in claim 8, wherein comprising performing a final quality check of the water and redirecting the water back to the raw water inlet (102) when measured water quality parameters do not fall within acceptable drinking water parameters.

Dated this 10th day of July, 2026



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ABSTRACT

AUTOMATIC WATER PURIFICATION SYSTEM AND METHOD FOR PURIFYING WATER FROM MULTIPLE NATURAL WATER SOURCES

5 The automatic water purification system (100) comprises a raw water inlet (102) to receive natural
water, a screening unit (104) to remove suspended impurities, and a plurality of sensors (106) that
measure pH, total dissolved solids, and turbidity. A microprocessor-based controller (108) processes
this sensor data to classify the water type and activate a corresponding purification sequence through
a plurality of purification stages (110) before delivery via a purified water outlet (112). Additionally,
the system features an electro-mineral conditioning module (114) with low-voltage DC-energized iron
10 (116) and silver (118) electrodes for mineral enrichment and antibacterial action. It also comprises a
pH stabilization module (120) using a calcium-magnesium alloy (122) for chemical-free potable pH
adjustment, and a salt removal module (124) utilizing a porous ceramic disk (126) with reverse osmosis
membranes (128) attached to its upper and lower surfaces.

(FIGURE 1)

15